21BCS028

Time: 1 hour | B.Tech V Semester, Sessional Test 1, 2023-24 | Max Marks: 15

Object Oriented Programming Systems (OOPS), CEN - 504

Q1 (1+1+3) 5 Marks Attempt ALL questions. **CO1**

(a) What are two **major** reasons to study object oriented design?

(b) Explain the different uses of the **const** keyword in the following declaration:

```
Point midPoint(const Point & p) const;
```

(c) Define a class **Point** with two double precision attributes **_x** and **_y**. Provide a default constructor, a parameterized constructor, a copy constructor, a default destructor and a **count** variable that keeps track of the number of instantiated objects present in the program. Provide the class definition and implementation.

Q2, (2.5+2.5) 5 Marks **CO1, CO2**

(a) Define a class **TempC** with a single **float** attribute that stores the temperature in degree Celsius and a class **TempF** with a single **float** attribute that stores the temperature in degree Fahrenheit. Declare and define a conversion operator that converts an object from **TempF** to **TempC**. (Hint: $(32°F - 32) \times 5/9 = 0°C$)

(b) Define a class **StrReplicate**. Include one data member which is a string. Write the complete definition of an overloaded **++ operator** (prefix notation) that works with the class **StrReplicate** and has the effect of appending another copy of the string to the end of **this** one.

Q3, (2+3) 5 Marks **CO2**

(a) Overload the stream insertion **operator <<** for the **TempC** class from Q2(a) that prints "X degrees Celsius" to the console output. Assume that the necessary constructors are present, what will be the output of the following code? Explain.

```
TempF summer(118); cout << summer;
```

(b) Define a class **Feature** that stores a 256 dimensional feature vector as a type **int** array. In the default constructor, initialize the vector to random values (Hint: **#include <cstdlib>, int rand()**). Declare and define an overloaded subscript operator [] that returns the ith element of the vector and can be used on LHS as well as RHS. Make a dataset (array) of 10,000 Feature objects and compute the centroid of the data.